2767-Machine Learning-2425_T3

Air Pollution Forecasting Using Machine Learning

Group T Members:

Sofia Pignataro 67523

Felix Otter 65284

Davide Dal Cero 64018

Francisco Manuel Nunes Mendes da Costa 67754

Moritz Müller 67194

Léon Philipp Klingele 67163

Date: March 23, 2025

Table of Contents

1 Introduction and Business Case **1**
1.1 Business Case 1

2 Exploratory Data Analysis (EDA) **1**

3 Methodology **2**
3.1 Feature Engineering 2
3.2 Feature Transformation 3
3.3 Model Selection 3
3.4 Hyperparameter Tuning 4

4 Model Evaluation **4**
4.1 Results 4
4.2 Explainability 5

5 Business Implications and Future Work **5**

A Appendix **6**

References **12**

1 Introduction and Business Case

Air pollution has become a critical environmental and public health issue worldwide, accounting for about 9 million deaths per year (1). Pollution levels are influenced by several factors, including human activities, industrial emissions, and meteorological conditions.
While many studies have focused on human-driven sources of pollution, there is growing evidence that atmospheric weather conditions significantly affect pollution levels (2), making it an essential area of study for improving environmental monitoring and management.
With this project, we aim to predict air pollution levels by incorporating meteorological variables, alongside historical pollution data, using machine learning techniques. To achieve this, we selected the dataset *"Air Pollution Forecasting – LSTM Multivariate"* from Kaggle. It contains hourly records of weather conditions and pollution levels over a five-year period, collected at the U.S. Embassy in Beijing, China.
For this project the pollution indicator, the "target variable" of the model, is PM2.5 concentration. It is a key indicator of air quality, measuring the concentration of fine particulate matter with a diameter of 2.5 micrometers or smaller. PM2.5 is a critical pollutant because it can penetrate deep into the respiratory system, causing serious health issues such as respiratory and cardiovascular diseases (4).

1.1 Business Case

As an American consulting firm contracted by the Municipality of Beijing, our objective is to analyze this dataset and design a machine learning pipeline and predictive model that forecasts air pollution levels on an hourly basis. The model should later power a service (such as a mobile application or SMS notifications) that will provide hourly air quality forecasts to Beijing residents.
The goal is to alert citizens when poor air quality is expected, allowing them to take preventive actions such as wearing masks or reducing outdoor activities, especially for individuals with respiratory conditions like asthma. On the other hand, the service will notify residents when air quality is expected to be good, encouraging outdoor activities and and improving awareness about current pollution.

2 Exploratory Data Analysis (EDA)

The dataset consists of hourly measurements of key environmental and meteorological factors, with the target variable being pollution (*PM2.5 concentration*). The explanatory variables include temperature, dew point, atmospheric pressure, wind speed, wind direction, snow, and rain. The objective of the exploratory data analysis (EDA) was to examine the distributions of these variables, detect potential anomalies, and identify relationships between pollution and meteorological factors. The dataset contains both numerical variables (pollution, temperature, dew, pressure, wind speed, rain, and snow) and a categorical variable (wind direction). No missing values or duplicate values were found in the dataset.
Histograms were generated to examine the distribution of environmental variables, revealing distinct patterns across the features. The target variable, pollution (*PM2.5*), exhibited a right-skewed distribution, with most values ranging between 0 and 300, but occasional peaks exceeding 600, indicating extreme pollution events. A similar right-skewness was observed in rain, wind speed, and snow, where the default state is the absence of these conditions, leading to a concentration of values around zero. However, occasional higher values were present, and as expected, no negative values appeared due to their natural constraints. In contrast, variables such as temperature, pressure, and dew point followed more balanced distributions. Based on these findings, section 3.2 explains how we applied feature transformation.
The correlation analysis showed that pollution concentration had a weak negative correlation with temperature (-0.091) and pressure (-0.046), suggesting that pollution levels tend to be slightly higher in colder and lower-pressure conditions. Additionally, wind speed (-0.23) had a weak negative correlation with pollution, indicating that stronger winds may contribute to pollution dispersion.
Among other variables, temperature and pressure had a strong inverse relationship (-0.83), reinforcing that higher temperatures typically coincide with lower pressure. Dew point was strongly correlated

with temperature (0.82), confirming that warmer air holds more moisture. Rain and snow showed no strong correlation with pollution, suggesting that precipitation may have a limited effect on air quality in this dataset.
Time-series analysis revealed the presence of seasonal patterns across environmental variables. Pollution levels exhibited frequent spikes exceeding 800, particularly during the winter months of 2010, 2012, and 2013, indicating clear seasonal peaks. Dew point and temperature followed strong cyclical trends, with temperatures rising to 20-30°C in summer and dropping as low as -30°C in winter, as expected. Conversely, pressure fluctuated between 990 and 1040 hPa, following an inverse relationship with temperature, where higher pressure was more common in winter and lower pressure in summer. Wind speed did not exhibit a distinct seasonal trend. Rainfall followed a different seasonal pattern, with sporadic peaks occurring during summer monsoon seasons, particularly in 2010, 2012, and 2014, with values reaching 35 mm. These findings confirm strong seasonal dependencies in most meteorological variables, with pollution levels showing a notable increase during winter months.
Finally, for the outlier detection, using the z-score method, we identified extreme pollution spikes above 400, especially during winter months. These data points likely represent real environmental phenomena rather than errors or noise, and removing them might result in a model that underestimates high-pollution events, which are critical for public health warnings.

3 Methodology

Before developing the final model, we explored different approaches to determine the most effective way to predict air pollution levels. The initial attempt involved aggregating the data at a daily level and predicting pollution categories based on daily summaries, including mean pollution levels, daily temperature ranges, and wind conditions. However, this approach resulted in poor model performance, achieving only 59% accuracy. The primary limitation was the loss of critical temporal dependencies, as air pollution fluctuations occur within shorter time frames and are highly influenced by hourly meteorological changes, wind shifts, and local emissions.
This underperformance indicated underfitting, as the model lacked sufficient complexity to capture the intricate patterns that characterize pollution trends. Given these limitations, a time-series approach at an hourly level was pursued to preserve fine-grained temporal patterns and better capture short-term pollution fluctuations. This decision was validated by a significant improvement in model performance, demonstrating the importance of maintaining high-resolution data in pollution forecasting.

3.1 Feature Engineering

Feature engineering is a critical step in the machine learning process and was particularly important for this time-series dataset. The objective was to create additional features that could help the model capture temporal dependencies and significant patterns in the data. The key feature engineering steps included:

- **Pollution Lag Variables:** To capture the temporal dependence of pollution levels, we created lag features. Specifically, we added a 24-hour lag (*pollution_lag_24*) to incorporate the pollution from the day before at the same hour.

- **Rolling Statistics:** A 24-hour rolling mean (*pollution_rolling_mean_24*) of pollution was computed to smooth short-term fluctuations and capture the broader pollution trend over the past day.

- **Wind Vector Components:** The wind direction (*wind_dir*) was converted into radians and used to derive wind vector components, *wind_x* and *wind_y*. These transformations allowed the model to interpret wind direction numerically, enabling a more unified representation across all observations.

- **High and Low Temperature Flags:** Binary features were introduced to indicate extreme temperature conditions: a *high_temp* flag for temperatures above 30°C and a *low_temp* flag for

temperatures below 0°C. According to a report from the World Resources Institute (3), extreme temperatures can significantly influence pollution levels, making these indicators valuable for prediction.

- **Time-Based Features:** To incorporate temporal dependencies, the day of the week (DOW) was extracted from the date feature. To preserve its cyclic nature, we applied cyclical encoding, transforming the values using sine and cosine functions. This ensures that adjacent days (e.g., Sunday and Monday) remain close in the transformed feature space, allowing the model to better recognize weekly patterns.

- **Seasonal Features:** The month of the year was used to assign a seasonal category to each observation: 1 for Spring, 2 for Summer, 3 for Fall, and 4 for Winter. Given that pollution levels exhibit seasonal trends, incorporating this feature helps capture long-term variations in air quality.

3.2 Feature Transformation

After the feature engineering, we applied appropriate transformations to ensure consistency across variables and enhance model performance.
Right-skewed features, such as pollution and its lagged variables, were log-transformed to reduce skewness and improve model stability. Continuous variables, including temperature, dew point, pressure, and wind vector components, were standardized to ensure that all features contributed equally to the model.
Rain and snow precipitation, which were highly skewed with mostly zero values and demonstrated no strong correlation with the target variable, were converted into binary features, indicating whether or not precipitation occurred in a given hour.
Since the goal of the project is to predict pollution levels, the pollution values were categorized based on percentiles to facilitate multi-class classification. We applied three pollution levels:

- **Low Pollution:** Pollution values less than or equal to the 25th percentile.

- **Medium Pollution:** Pollution values between the 25th and 75th percentiles.

- **High Pollution:** Pollution values greater than or equal to the 75th percentile.

This classification framework allows the Municipality of Beijing to understand when pollution levels reach the *danger zone* (High), enabling timely interventions and precautionary measures to safeguard public health.

3.3 Model Selection

Among the various models considered, we selected the Random Forest as the primary model due to its robustness against overfitting while maintaining interpretability in a business context. While deep learning models offer powerful performance, Random Forest provides clear insights into feature importance and decision-making processes. Previous studies (5) highlight the effectiveness of Random Forest Regression in capturing complex relationships between meteorological variables and air quality levels.
While Random Forest was ultimately chosen for deployment, other models were explored during the selection process. Logistic Regression was tested as a baseline model, providing a reference point for performance evaluation. Additionally, Gradient Boosting Machines (GBM) were considered, leveraging an ensemble learning approach that builds trees sequentially, correcting errors from previous iterations.
Through this comparative analysis, Random Forest emerged as the best-performing model, offering the optimal balance of accuracy, interpretability, and recall for the critical *High Pollution* class. This selection ensures that the model not only delivers reliable predictions, but also remains accessible for business decision-makers and policymakers who require an intuitive understanding of its outputs.

3.4 Hyperparameter Tuning

One of the key steps in improving model performance is the careful selection of hyperparameters. The predictive capability of Random Forest is highly dependent on hyperparameter choices, which influence the model's complexity, generalization ability, and computational efficiency. To optimize these parameters, we employed `RandomizedSearchCV`, a technique that efficiently explores a range of hyperparameter values using cross-validation. Unlike `GridSearchCV`, which exhaustively evaluates all possible combinations, `RandomizedSearchCV` allows for a broader search space while maintaining computational efficiency.

Since the target variable was categorized into three classes (*Low, Medium, High*), we utilized the `predict_proba()` method in Random Forest to estimate the probability distribution across classes. By default, the model assigns each observation to the class with the highest probability. However, to enhance the recall for the *High Pollution* class, we adjusted the decision threshold.

Maximizing recall for the high pollution class was prioritized due to the critical implications of misclassification. In air quality monitoring, a false negative—where the model predicts low or medium pollution when actual pollution levels are high—poses a significant public health risk. Conversely, a false positive, while suboptimal, is less detrimental. Adjusting the threshold allowed us to improve the model's ability to detect high pollution episodes, ensuring timely warnings and preventive actions.

4 Model Evaluation

To assess the performance of our model, we employed multiple evaluation metrics to ensure a comprehensive understanding of its predictive capabilities. The primary metrics used were:

- **Accuracy**: The proportion of correctly classified instances across all classes.
- **Precision, Recall, and F1-Score**: These metrics were computed for each pollution category to evaluate the balance between false positives and false negatives. Precision measures the proportion of correctly predicted instances out of all predicted instances for a given class, while recall quantifies the proportion of actual instances correctly identified. The F1-score provides a harmonic mean between precision and recall, ensuring a balanced assessment.
- **ROC AUC (Receiver Operating Characteristic - Area Under the Curve)**: The AUC measures the model's ability to distinguish between pairs of classes. A high AUC indicates strong performance in ranking instances correctly, even across varying decision thresholds. We evaluate AUC for each class pair (One-vs-One), to understand at which boundary (low vs medium class or medium vs high class) the model needs improvement.

4.1 Results

To evaluate model performance, multiple classification algorithms were tested and compared. Logistic Regression was initially used as a baseline model, achieving an accuracy of 74.3% 1. While it demonstrated reasonable predictive power, particularly for the medium pollution class, its recall for the high pollution class remained lower at 71.4%. Given the critical importance of correctly identifying high pollution levels, this model was deemed insufficient for deployment.

XGBoost showed improvements over Logistic Regression, attaining an overall accuracy of 78.8% 4. Notably, it provided a higher recall of 85.8% for the high pollution class, suggesting its enhanced ability to detect severe pollution events. However, despite its gains, it still underperformed compared to the Random Forest model, which was ultimately selected for deployment.

The Random Forest classifier achieved an accuracy of 81.7% 2, with a recall of 87.9% for the high pollution class. Adjusting the classification threshold slightly reduced the overall accuracy by 0.8%, but it resulted in a 10% improvement in detecting high pollution cases. Given the project's objective of providing an early-warning system for hazardous air quality conditions, this trade-off was considered acceptable, as prioritizing recall for high pollution levels ensures that critical warnings are issued effectively. The ROC AUC analysis 10 further supports the model's strong performance, with AUC

values of 0.95 for low vs high pollution and 0.91 for low vs medium pollution, indicating excellent class separation. The slightly lower AUC of 0.89 for medium vs high pollution suggests room for improvement, but overall, the model demonstrates robust discrimination across all class pairs, reinforcing its ability to detect high pollution cases effectively.
To assess the stability and generalization ability of the final model, five-fold cross-validation was performed. The model yielded a mean cross-validation accuracy of 81.1%, with a standard deviation of 0.38%. These results confirm the model's consistency and robustness across different data subsets, reinforcing its suitability for real-world application.

4.2 Explainability

To better understand the key drivers of pollution prediction, we analyzed the feature importances in the final Random Forest model . The results highlighted that pollution trends and wind-related factors were the most influential predictors. The 24-hour rolling mean of pollution was the most important feature (36.4%), confirming the strong temporal dependency of pollution levels and the relevance of recent pollution trends in forecasting future conditions.
Wind components (*wind_x* and *wind_y*) accounted for 11.7% and 10% of the importance, respectively, emphasizing the role of wind direction and intensity in pollution dispersion. Additionally, pollution lag variables (*pollution_lag_24*) contributed 11%, further demonstrating that pollution exhibits strong temporal dependencies. These results validate the effectiveness of our feature engineering approach, as the most influential predictors—pollution trends, wind components, and lag variables—were specifically designed to capture the temporal and meteorological patterns affecting pollution levels.
Other meteorological factors also played a role in pollution prediction. Dew point (9.7%) and temperature (6.1%) emerged as relevant variables, reinforcing the known influence of atmospheric conditions on pollution fluctuations. Time-related features, such as the day of the week and seasonal indicators, had lower importance but still contributed to capturing long-term seasonal variations in pollution patterns.

5 Business Implications and Future Work

The deployment of this model offers a practical and data-driven approach to air quality monitoring, allowing the Municipality of Beijing to proactively address pollution risks. By prioritizing recall for high pollution levels, the model ensures that critical pollution events are detected with high reliability, reducing the likelihood of failing to warn the public about hazardous air quality conditions. This is particularly important for safeguarding public health, as timely warnings enable citizens to take preventive measures, such as limiting outdoor activities or wearing protective masks.
From a business perspective, Random Forest was selected as the final model due to its balance between accuracy, interpretability, and computational efficiency, making it a robust solution for operational implementation. Unlike deep learning models, which are often considered black-box methods, Random Forest provides clear feature importance rankings, enabling decision-makers to understand the key drivers of pollution and adjust policies accordingly. This transparency is particularly valuable for public agencies and policymakers, as it facilitates data-driven interventions, such as traffic restrictions or temporary industrial activity limitations during periods of high pollution.
This model can also serve as the foundation for an automated alert system, where authorities can trigger pollution warnings based on predicted probability thresholds. Such a system could be integrated into mobile applications or SMS notification services to inform residents about expected pollution levels and necessary precautions.
Future improvements could focus on incorporating additional external factors that influence pollution levels. Potential enhancements include integrating real-time vehicular traffic data, industrial activity levels, and satellite-based air quality readings to refine the model's predictions. Additionally, expanding the dataset with air quality readings from multiple locations could further generalize the model's applicability. These improvements would strengthen the model's predictive capabilities, ultimately contributing to more effective air quality management and public health protection.

A Appendix

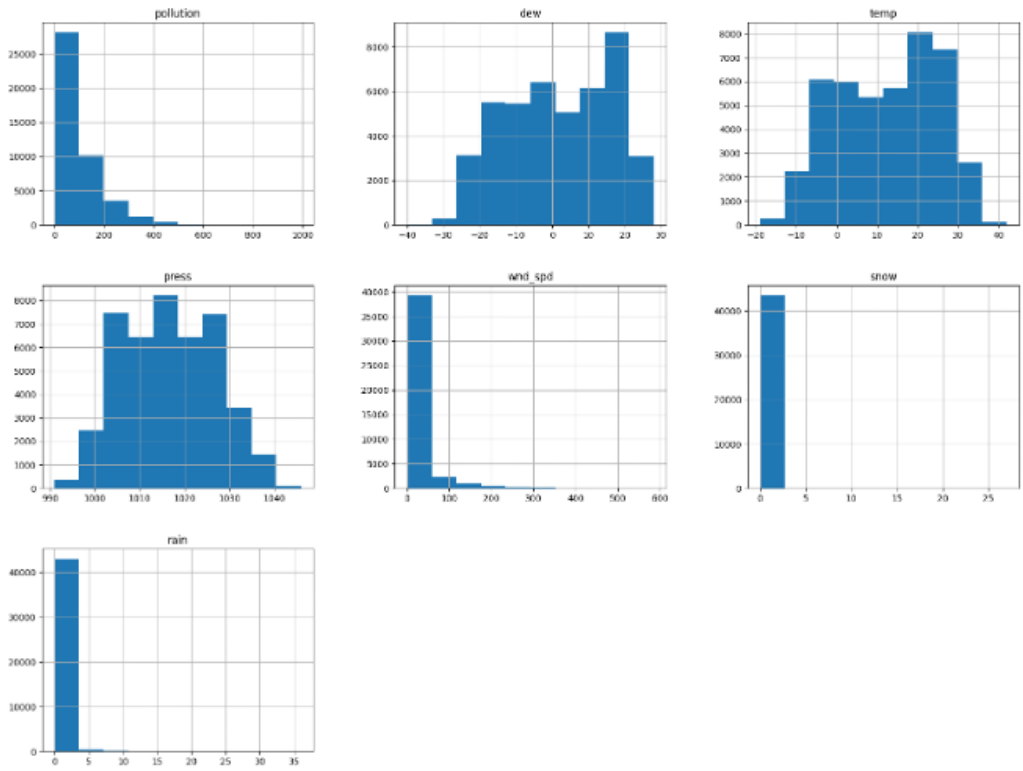

Figure 1: Distribution and variability of key environmental variables

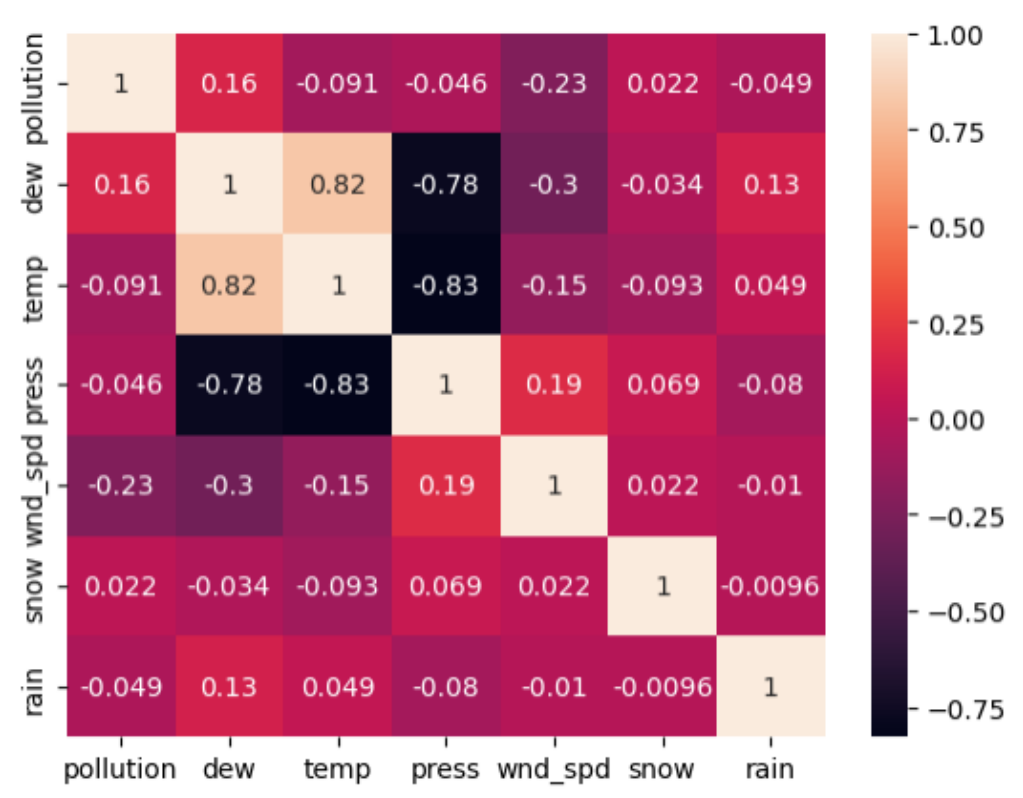

Figure 2: Relationships between key variables

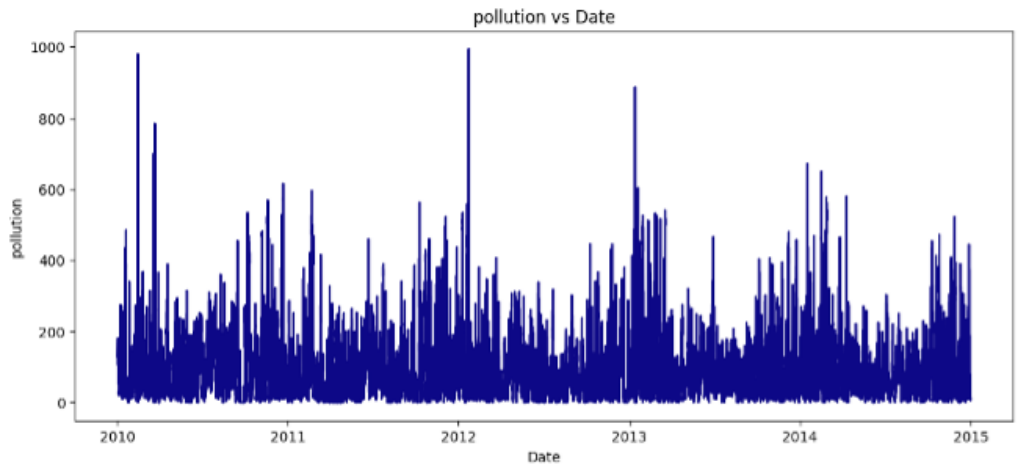

Figure 3: Seasonality of Pollution

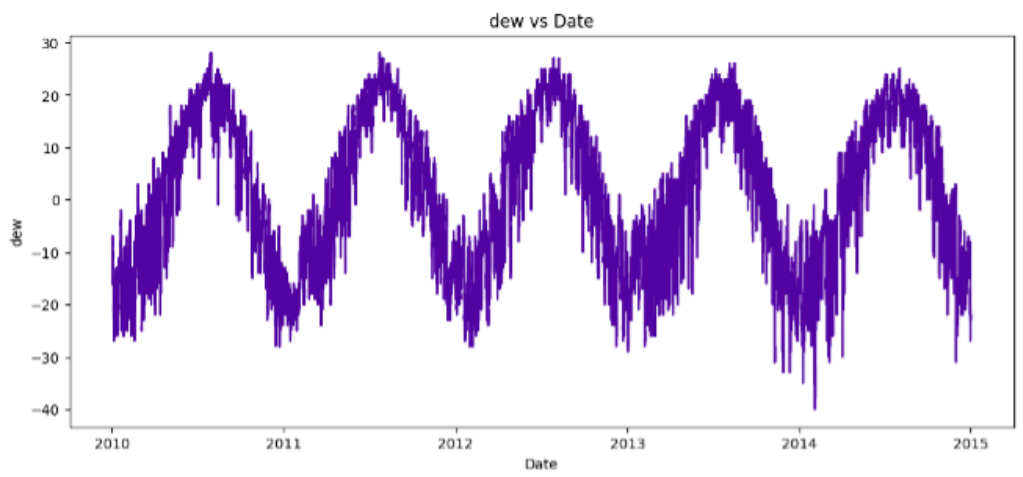

Figure 4: Seasonality of Dew

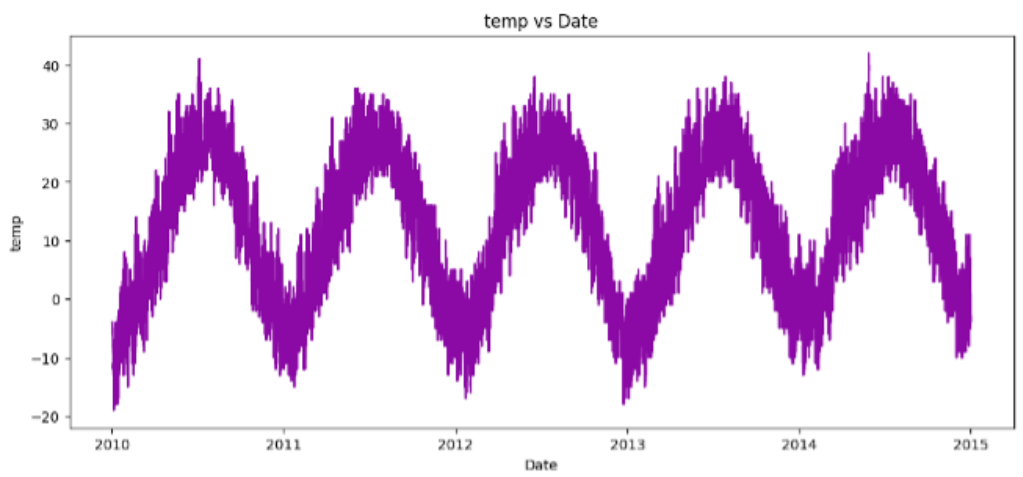

Figure 5: Seasonality of Temperature

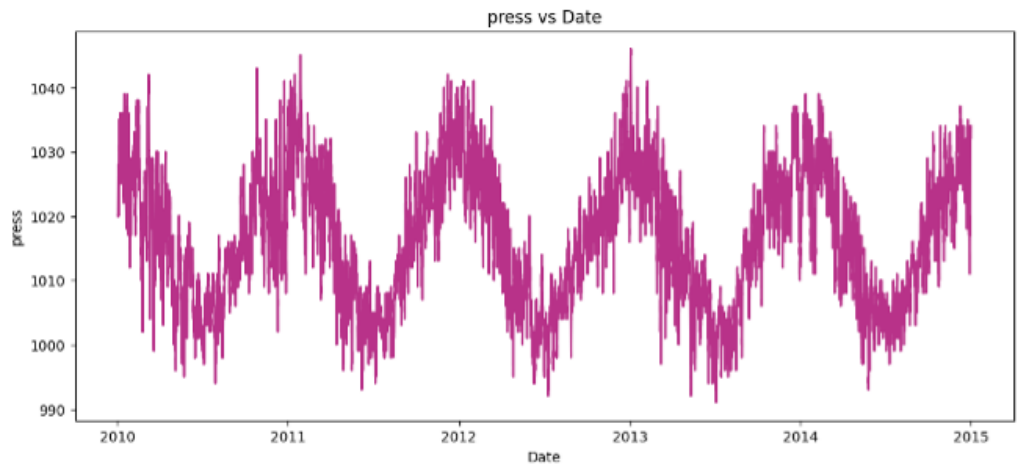

Figure 6: Seasonality of Air Pressure

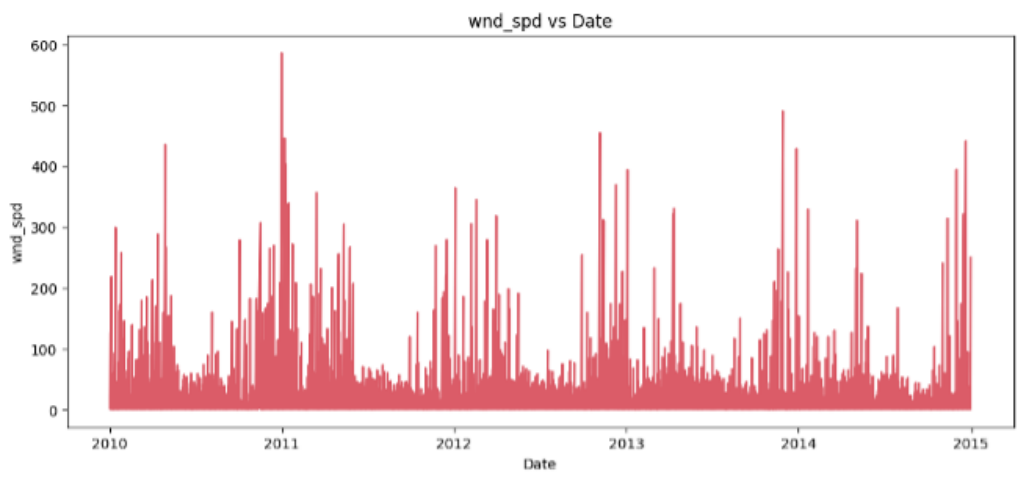

Figure 7: Seasonality of Wind Speed

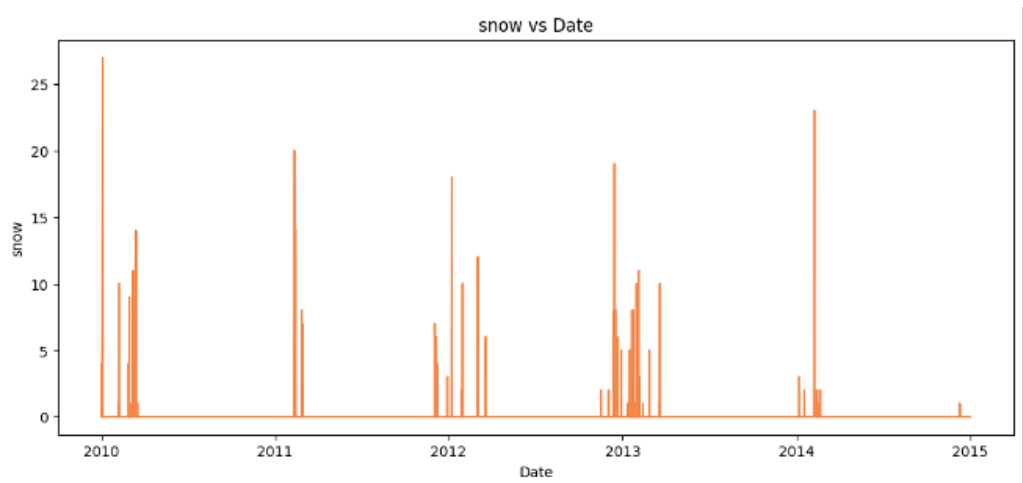

Figure 8: Seasonality of Snow Fall

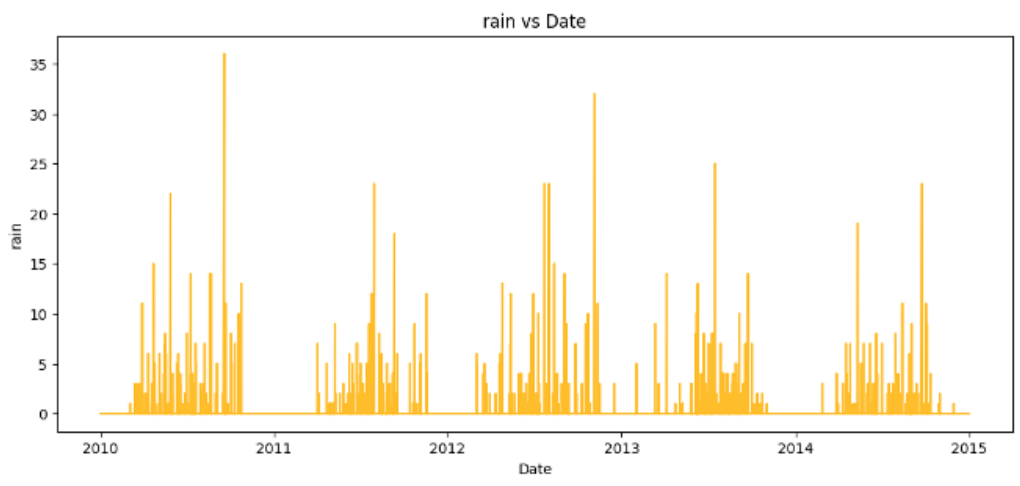

Figure 9: Seasonality of Rain

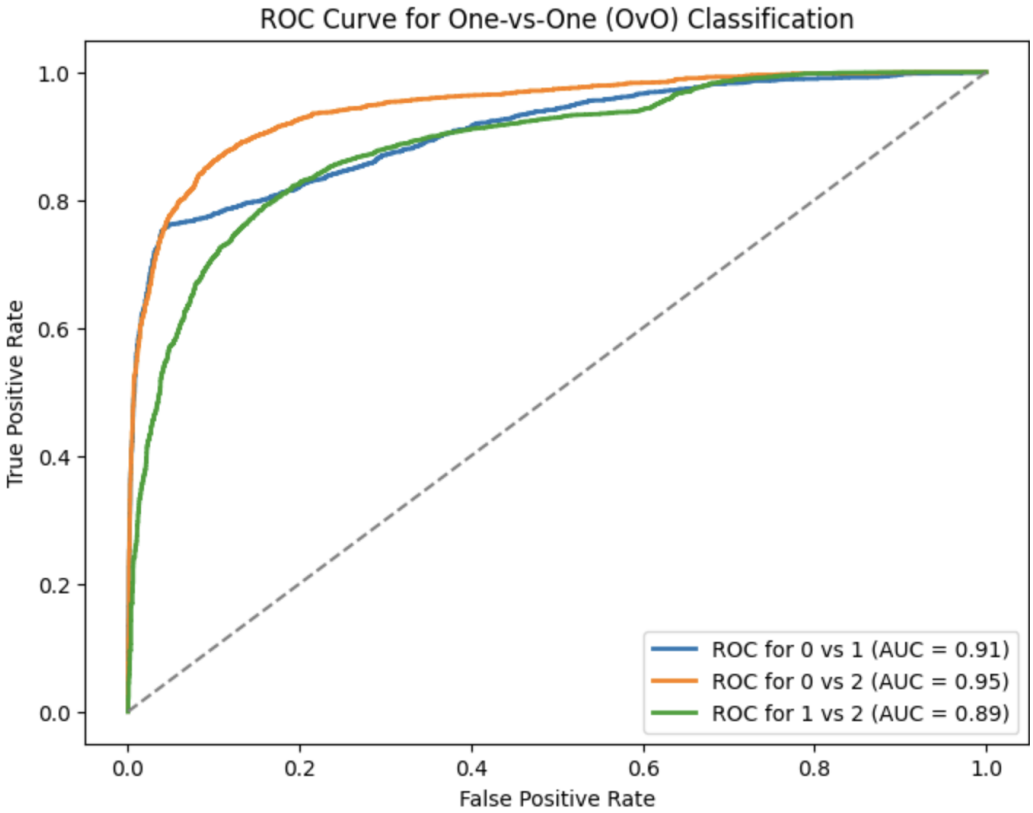

Figure 10: One-vs-one ROC-AUC

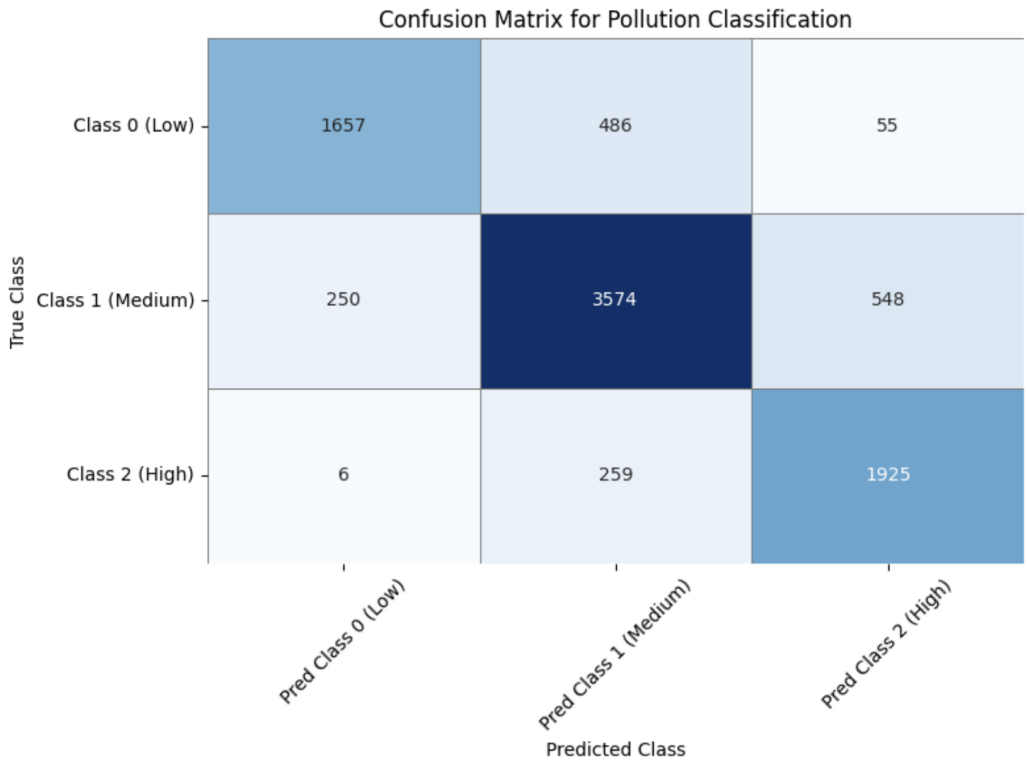

Figure 11: Confusion Matrix with Random Forest and threshold adjusted

	Feature	Importance
9	pollution_rolling_mean_24	0.363654
11	wind_y	0.116869
8	pollution_lag_24	0.110379
10	wind_x	0.099902
0	dew	0.097517
1	temp	0.061919
2	press	0.055813
12	DOW_sin	0.025629
13	DOW_cos	0.018925
14	Season_1	0.008855
4	rain	0.007321
15	Season_2	0.007146
5	is_weekend	0.006587
17	Season_4	0.006318
7	low_temp	0.004552
16	Season_3	0.004302
6	high_temp	0.002584
3	snow	0.001728

Figure 12: Feature Importance for every attribute included in the final model

Logistic Regression (Hourly Aggregated Data)

Metric	Value
Accuracy	0.743151
Recall (Class 2)	0.713699

Table 1: Logistic Regression Results

Random Forest Classifier (Hourly Aggregated Data - With and Without Adjusted Threshold)

Metric	Value
Accuracy	0.816895
Precision (Class 0)	0.866179
Recall (Class 0)	0.753867
F1-score (Class 0)	0.806130
Precision (Class 1)	0.827506
Recall (Class 1)	0.817475
F1-score (Class 1)	0.822460
Precision (Class 2)	0.761472
Recall (Class 2)	0.878995
F1-score (Class 2)	0.816024

Table 2: Random Forest Classifier (Adjusted Threshold 0.3) Results

Metric	Value
Accuracy	0.824658
Precision (Class 0)	0.865866
Recall (Class 0)	0.754777
F1-score (Class 0)	0.806514
Precision (Class 1)	0.798385
Recall (Class 1)	0.882205
F1-score (Class 1)	0.838205
Precision (Class 2)	0.848485
Recall (Class 2)	0.779909
F1-score (Class 2)	0.812753

Table 3: Random Forest Classifier (Without Adjusted Threshold) Results

XGBoost Classifier (Hourly Aggregated Data - Adjusted Threshold)

Metric	Value
Accuracy	0.788128
Precision (Class 0)	0.832909
Recall (Class 0)	0.743858
F1-score (Class 0)	0.785869
Precision (Class 1)	0.806521
Recall (Class 1)	0.775160
F1-score (Class 1)	0.790530
Precision (Class 2)	0.724470
Recall (Class 2)	0.858447
F1-score (Class 2)	0.785789

Table 4: XGBoost Classifier (Adjusted Threshold) Results

References

[1] World Health Organization, *Air Pollution and Health Effects*, 2023.

[2] U.S. Environmental Protection Agency, *Understanding Air Quality Trends*, 2022.

[3] World Resources Institute, *What Happens When Extreme Heat and Air Pollution Collide*, 2024.

[4] World Resources Institute, *The Impact of PM2.5 on Human Health*, 2021.

[5] Yu et. al, *RAQ–A Random Forest Approach for Predicting Air Quality in Urban Sensing Systems*, 2016.